

Kunjal Panchal

📞 (413) 210 9198 | ✉ kpanchal@umass.edu | 🏠 astuary.github.io/Kunjal/ | 🐙 github.com/astuary | 🔗 linkedin.com/in/kunjal-panchal

Skills

Research	Resource-constrained Learning, Agentic Harnesses, Personalization, On-device Inference, Edge AI.
Techniques	Model Compression, Quantization, Memory- and Compute-Efficient Training/Inference, Distributed & Parallel ML Algorithms, Test-Time Adaptation, Optimization.
Frameworks	PyTorch, Hugging Face, Flower (Federated Learning Framework), Torch Distributed, PyTorch Mobile, ExecuTorch.

Education

University of Massachusetts Amherst

Doctor of Philosophy in Computer Science (3.7/4.0 GPA)

Amherst, MA

Sep 2021 - Expected May 2026

- Advisor: Dr. Hui Guan.
- James Kurose Scholar (scholarship given for systems in machine learning project, Spring 2022).
- Jumpstart Fellow (fellowship given to top 5 research proposals by new PhD students, Fall 2021).

University of Massachusetts Amherst

Master of Science in Computer Science Research Track (3.6/4.0 GPA)

Amherst, MA

Sep 2019 - May 2021

- Advisor: Dr. Adam O'Neill.
- Research Area: Relaxed Cryptography for Digital Signatures and Message Authentication Codes.
- **Courses:** Machine Learning, Computer Vision, Natural Language Processing, Reinforcement Learning, Robotics, Optimization in Computer Science, Advanced Algorithms, Modern Computer Architecture, Advanced Cryptography.

Work Experience

Adobe Research

Research Scientist/Engineer

San Jose, CA

May 2026 - Current (Full-time)

- Conducting research to design, develop, and optimize machine learning and AI models, for scalable, real-world applications.
- Developing efficient on-device AI algorithms for vision and multimodal models, enabling low-latency inference on resource-constrained mobile and edge devices.

Adobe Research

Research Scientist/Engineer Intern

San Jose, CA

May 2025 - Aug 2025 (Full-time);

Aug 2025 - Nov 2025 (Part-time)

- Designed an LLM planning algorithm for design personalization, introducing a novel **belief-shift scoring rule** and preference extractor.
- Designs (ads/posters) generated by the proposed system were chosen by **68% of designers as top-1 (93% top-2)** and rated **4.31/5**, outperforming baselines and aligning with AI-judge rankings.

Research

Mosaic: Runtime-Efficient Multi-Agent Embodied Planning

Kunjal Panchal, Saayan Mitra, Sunav Choudhary, Victor Bursztyn, Somdeb Sarkhel, Hui Guan

Published @ ICML, 2026.

Memory Savings at What Cost? A Study of Alternatives to Backpropagation

Kunjal Panchal, Sunav Choudhary, Yuriy Brun, and Hui Guan

Published @ ICML, 2026.

ATOM: Efficient On-Device Video-Language Pipelines Through Modular Reuse

Kunjal Panchal, Saayan Mitra, Haoliang Wang, Somdeb Sarkhel, Ishita Dasgupta, Gang Wu, Hui Guan

Published @ ACM MMSys, 2026.

Thinking Forward: Memory-Efficient Federated Finetuning of Language Models

Kunjal Panchal, Nisarg Parikh, Sunav Choudhary, Yuriy Brun, and Hui Guan

Published @ NeurIPS, 2024.

Flash: Concept Drift Adaptation in Federated Learning

Kunjal Panchal, Sunav Choudhary, Koyel Mukharjee, Subrata Mitra, Somdeb Sarkhel, Saayan Mitra, and Hui Guan

Published @ ICML, 2023.

Flow: Per-instance Personalized Federated Learning

Kunjal Panchal, Sunav Choudhary, Nisarg Parikh, Lijun Zhang, and Hui Guan

Published @ NeurIPS, 2023; Preliminary Presentation @ CrossFL, MLSys 2022.

Leadership / Volunteering

- Current **Peer Reviewer**, NeurIPS '26/'25 (Top Reviewer)/'24, ICML '26 (Gold Reviewer)/'25, ICLR '27/'26/'25, TMLR '26, TNNLS '26, AISTATS '26/'25, AAAI '27/'26/'25/'24, ACM MM '25, TSE '25
- Oct 2025 **Workshop Organizer**, Short-Form Video Understanding @ ICCV 2025
- Oct 2024 **Poster Presenter**, UMass Amherst CS Department Undergrad Research Night
- Research Mentor for Undergraduates**, UMass Amherst CS department program to cultivate interest & understanding in research (Dec 2024, Jun 2024, Dec 2023, Jun 2023, Dec 2022)
- Applied Deep Learning Head Mentor**, Teaching applied deep learning to undergraduates at SureStart (Volunteer position) (Jun 2024, Jun 2023, Jan 2023, Jun 2022)
- Nov 2023 **Panelist and Poster Presenter**, UMass Amherst CS Department Undergrad Research Night
- Jan 2022 **Coding Gym Leader**, SureStart winter bootcamp to teach coding interview strategies
- Oct 2021 **PhD Applicant Support Program**, Mentoring prospective PhD applicants
- Mar 2021 **Machine Learning Mentor**, Virtual AI Learning Program hosted by SureStart
- Aug 2020 **Emotion AI Program Mentor**, EMPATH Program hosted by Affectiva
- Dec 2019 **Campus Leader**, Google Developer Students Club India

Achievements

- 2022 **James Kurose Scholarship**, Manning College of Info and Comp Sci, UMass Amherst
- 2021 **CICS Jumpstart Fellowship**, College of Info and Comp Sci, UMass Amherst
- 2019 **Gold Medalist**, The Maharaja Sayajirao University of Baroda, B.Engg. in Computer Science
- 2019 **Student of the Year**, The Maharaja Sayajirao University of Baroda, B.Engg. in Computer Science
- National Talent Search Examination**, Top 100 in Science and Mathematics in India
- All India Essay Writing Event**, Honorable Mention in a state-level essay competition
- Community Science Center**, Winner of Conmat Cosmopolitan Tree Garden Award at state-level

Presentations

- Career Pathways Seminar Speaker *Spring 2025*
- Delivered an introductory talk on privacy-preserving machine learning and on-device inference to second-year undergraduates, covering industry applications and open research challenges.
- Voices of Data Science Poster Presenter *Spring 2023*
- Showcased "Flash: Concept Drift Adaptation in Federated Learning" (ICML '23) in an interdisciplinary poster session hosted across computer science, engineering and social/behavioral science departments. Winner of the poster presentation competition.
- Computer Science Research Night Poster Presenter *Fall 2022*
- Introduced my lab and research to undergraduate and graduate students looking to understand and participate in the ongoing research works.
- Cryptography Honors Seminar Speaker *Fall 2022*
- Discussed federated learning, differential privacy, applications, and why confidentiality of data is important in the world which is shifting towards data-rich artificial intelligence.
- AI4ALL Summer Program Speaker *Summer 2021*
- Presented detailed pointers on how to read, understand, write research papers in AI and ML.
 - Explained how to figure out unsolved yet solve-able problems, conduct research through creative solutions, evaluate results derived of the proposed approach, and discussed ethics and biases in AI.
 - Encouraged 20+ undergraduate students from Boston University, Columbia University, and University of California Berkeley to pursue artificial intelligence research.